Robust quantum gates for stochastic time-varying noise

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How to effectively construct robust quantum gates for time-dependent noise is a very important but still challenging problem. Here we develop a systematic method to find pulses for quantum gate operations robust against both low- and high-frequency (comparable to the qubit transition frequency) stochastic time-varying noise. Our approach, taking into account the noise properties of quantum computing systems, can output single smooth-shaped pulses in the presence of multiple types of noise. Furthermore, our method can be applied to different system models and noise models and will make essential steps toward constructing high-fidelity and robust quantum gates for fault-tolerant quantum computation. Finally, we discuss and compare the gate operation performance by our method with that by the filter-transfer-function method.

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I. INTRODUCTION

In realistic practical quantum computations, a set of high-fidelity universal quantum gates robust against noise in the qubit or quantum environment is required. Constructing control pulses to operate quantum gates which meet this requirement is an important and timely issue. Quantum gates in open quantum systems have been investigated by various methods such as dynamical decoupling methods [1–6] and optimal control methods [7–10]. For classical noise, there are many robust control methods such as composite pulses [11–19], self-consistent pulses against the coupling and drift noise (SUPCODE) [20,21], the sampling-based learning control method [26,28], robust control methods [29,30], the analytical method [31], single-shot pulse method [32], optimal control methods [33–35], the unitary-based inverse engineering methods [36,37], and filter transfer function (FTF) methods [38–43]. However, most of these methods [11–34] assume the noise to be quasistatic, i.e., no time dependence within the gate operation time but can vary between gates. We call these the quasistatic noise (QSN) methods for the QSN assumption to be relevant [13]. The robustness of control pulses obtained by QSN methods against time-dependent noise, e.g., $1/f^\alpha$ noise [22,23,25,30], has been investigated, and it was found that they can still work well for relatively low-frequency non-Markovian noise (e.g., $1/f^\alpha$ noise with $\alpha \geq 1$).

Stochastic time-dependent noise is treated with the FTF method [38–43], in which the area of the filter transfer function in the frequency region, where the noise power spectral density (PSD) is nonnegligible, is minimized. However, in this approach only the the filter transfer function overlapping with the noise PSD in the [small] frequency region is considered, and the detailed information on the distribution of the noise PSD is not included in the optimization cost function. Here we present an optimal control method in the time domain by choosing the ensemble average gate infidelity as the cost function for optimization. As a result, the noise correlation function (CF) [or, equivalently, the detailed noise PSD distribution] appears explicitly in our chosen optimization cost function. Therefore our method can show a better control performance against noise in the general case. The [illegible] method is simple, and our method is not limited to particular control models, noise models, or noise types. For demonstration, we use a robust control method for classical noise in this paper, but our method can be readily generalized to the case of quantum noise by replacing the ensemble average for classical noise [44] with the trace over the degrees of freedom of the quantum noise environment [44]. In other words, our method can be applied to systems with both classical noise and quantum noise (see the Appendix).

II. ENSEMBLE AVERAGE FIDELITY AND OPTIMIZATION METHOD

We first introduce our robust control method here and then compare it with the QSN method and the FTF method. We consider a total Hamiltonian $H(t) = H_c(t) + H_N(t)$, where $H_c(t)$ is the ideal system Hamiltonian and $H_N(t)$ is the noise Hamiltonian. If noise is absent (i.e., $H_N(t) = 0$), then the propagator is $U_c(t) = \mathcal{T}_+ \exp[-i\int_0^t H_c(t')dt']$. Throughout the paper we set $\hbar = 1$, where $\mathcal{T}_+$ is the time-ordering operator. However, in reality, there may be many sources of noise present in the system, so $H_N(t) = \sum_\mu \beta_\mu(t) H_\mu(t)$, where $\beta_\mu(t)$ is the strength of the μth stochastic time-varying noise (STVN) and $H_\mu(t)$ is the corresponding system coupling operator. The propagator for a stochastic realization is then $U(t) = U_c(t)\mathcal{T}_+ \exp[-i\int_0^t \tilde{H}_N(t')dt']$. Here $\tilde{H}_N(t) = \sum_\mu \beta_\mu(t)\tilde{H}_\mu(t)$ is the noise Hamiltonian in the interaction picture transformed by $U_c(t)$ and $\tilde{H}_\mu(t) = U_c^\dagger(t) H_\mu(t) U_c(t)$. Suppose that U_T is our target gate and the gate operation time is t_g. The gate infidelity (error) $\mathcal{I}$ for an n-qubit gate can be defined as

$$\mathcal{I} = 1 - \frac{1}{d^2}|\mathbb{T}[U_T^\dagger U(t_g)]|^2, \qquad (1)$$

where $\mathbb{T}$ denotes a trace over the n-qubit system state space. If the noise strength is not too strong, we can expand the propagator $U(t_g)$ in terms of $\tilde{H}_N(t)$ by Dyson series [45] and put the form $U(t_g) = U_c(t_g)[I + \Phi_1 + \Phi_2 + \cdots]$, where the first two terms of Φ are $\Phi_1 = -i\int_0^{t_g}\tilde{H}_N(t)dt$ and $\Phi_2 = -\int_0^{t_g}dt_1\int_0^{t_1}dt_2\tilde{H}_N(t_1)\tilde{H}_N(t_2)$. Substituting the expanded $U(t_g)$ into $\mathcal{I}$ in Eq. (1), the expanded infidelity $\mathcal{I}$

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